CBSE Class 10 Mathematics — Circles

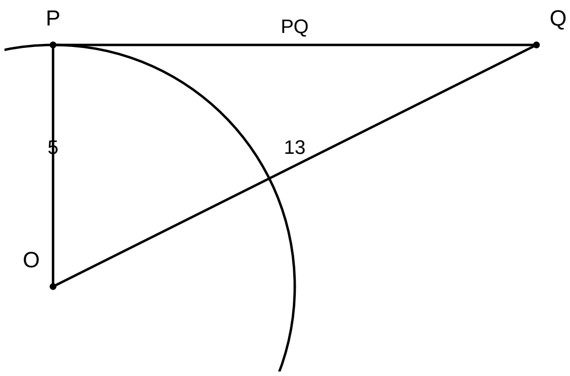
Time Allowed: 3 Hours **Maximum Marks: 80**

General Instructions:

1. This question paper contains 24 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 13$ cm. The length of PQ is:



(A) (A) 10 cm
(B) (B) 11 cm
(C) (C) 12 cm
(D) (D) 8 cm

2. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is equal to:

(A) (A) 50°
(B) (B) 60°
(C) (C) 70°
(D) (D) 80°

3. The length of a tangent from a point A at a distance of 5 cm from the centre of the circle is 4 cm. The radius of the circle is:

(A) (A) 2 cm
(B) (B) 3 cm

(C) (C) 4 cm

(D) (D) 5 cm

4. Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:

(A) (A) 6 cm

(B) (B) 7 cm

(C) (C) 8 cm

(D) (D) 10 cm

5. A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, then AD is equal to:

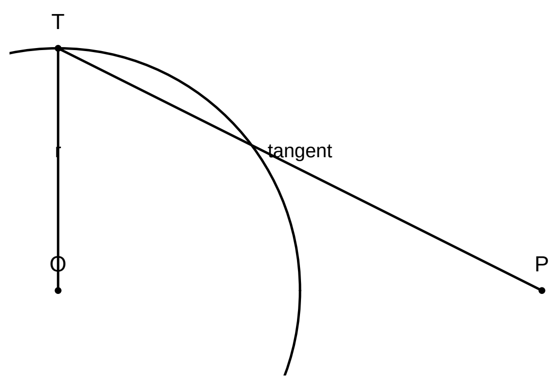
(A) (A) 2 cm

(B) (B) 4 cm

(C) (C) 5 cm

(D) (D) 3 cm

6. Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): A tangent to a circle is perpendicular to the radius through the point of contact.



(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

7. Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent to a circle intersects the circle at only one point.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

8. Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is $60°$, then $OP = 2r$. Reason (R): The line segment joining the center to the external point bisects the angle between the two tangents.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

9. Assertion (A): Two concentric circles have the same center. Reason (R): A chord of the larger circle which touches the smaller circle is bisected at the point of contact.

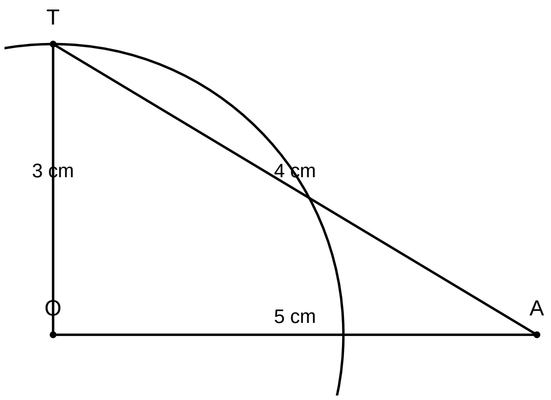
(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): The length of the tangent from a point A at a distance of 5 cm from the centre of the circle of radius 3 cm is 4 cm. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.



(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

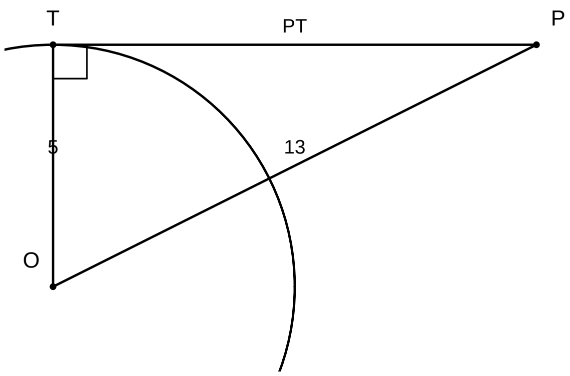
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

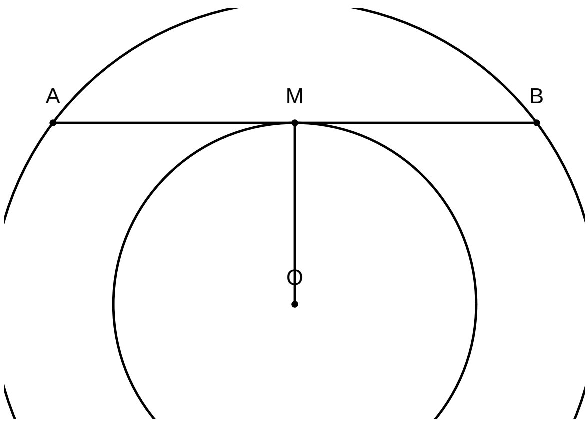
(D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. In a circle with centre O, a tangent PT is drawn from a point P outside the circle. If $OP = 13\ cm$ and the radius of the circle is $5\ cm$, find the length of the tangent PT.



2. Two concentric circles are of radii $5\ cm$ and $3\ cm$. Find the length of the chord of the larger circle which touches the smaller circle.



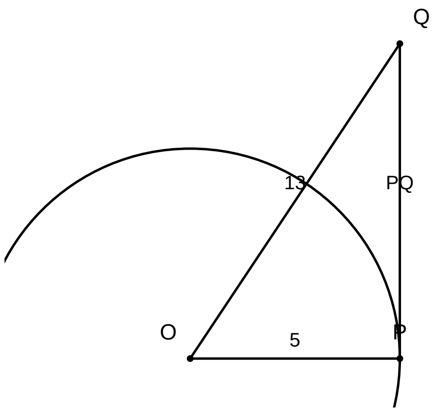
3. PA and PB are two tangents from a point P to a circle with centre O such that $\angle APB = 80°$. Find $\angle AOB$.

4. If a tangent PA and PB from a point P to a circle with centre O are inclined to each other at an angle of $70°$, then find $\angle POA$.

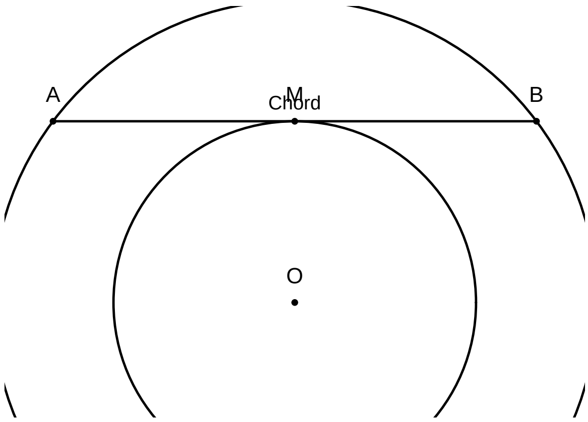
5. A point P is $26\ cm$ away from the centre O of a circle and the length of the tangent PT drawn from P to the circle is $10\ cm$. Find the radius of the circle.

Section D: Long Answer (4-5 marks each)

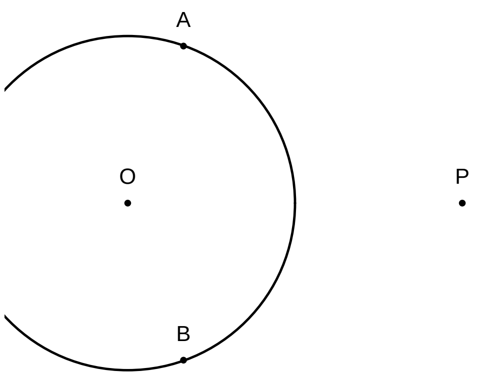
1. A tangent PQ at a point P of a circle of radius $5\ cm$ meets a line through the centre O at a point Q so that $OQ = 13\ cm$. Find the length of PQ.



2. Two concentric circles are of radii 5 *cm* and 3 *cm*. Find the length of the chord of the larger circle which touches the smaller circle.



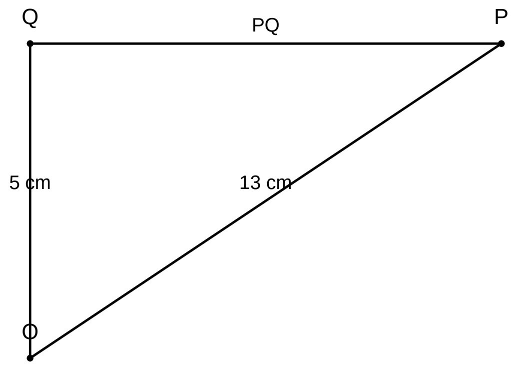
3. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at angle of $80°$, then find $\angle POA$.



4. Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

5. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

6. Prove that the lengths of tangents drawn from an external point to a circle are equal. Using this result, find the length of PQ if PQ is a tangent to a circle with centre O and radius 5 cm, where $OQ = 13$ cm and P is the external point.



7. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

8. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

9. If a triangle ABC is drawn to circumscribe a circle of radius 2 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 4 cm and 3 cm respectively, find the sides AB and AC.